

DESARROLLO EJERCICIO 1

Carnet: 00004624

A: 00

B: 24

Red base asignada: 10.24.0.0/24

Configuracion IP de cada PC

```
PC1> ip 10.24.0.2 255.255.255.192 10.24.0.1
Checking for duplicate address...
PC1 : 10.24.0.2 255.255.255.192 gateway 10.24.0.1
```

```
PC1> save
Saving startup configuration to startup.vpc
. done
```

```
PC1> show ip
```

```
NAME       : PC1[1]
IP/MASK     : 10.24.0.2/26
GATEWAY     : 10.24.0.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 10028
RHOST:PORT  : 127.0.0.1:10029
MTU         : 1500
```

```
PC1> █
```

```
PC2> ip 10.24.0.66 255.255.255.224 10.24.0.65
Checking for duplicate address...
PC1 : 10.24.0.66 255.255.255.224 gateway 10.24.0.65
```

```
PC2> save
Saving startup configuration to startup.vpc
. done
```

```
PC2> show ip
```

```
NAME       : PC2[1]
IP/MASK     : 10.24.0.66/27
GATEWAY     : 10.24.0.65
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 10030
RHOST:PORT  : 127.0.0.1:10031
MTU         : 1500
```

```
PC2> █
```

```
PC3> ip 10.24.0.98 255.255.255.240 10.24.0.97
Checking for duplicate address...
PC1 : 10.24.0.98 255.255.255.240 gateway 10.24.0.97

PC3> save
Saving startup configuration to startup.vpc
. done

PC3> show ip

NAME       : PC3[1]
IP/MASK    : 10.24.0.98/28
GATEWAY    : 10.24.0.97
DNS        :
MAC        : 00:50:79:66:68:02
LPORT      : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500

PC3> █
```

```
PC4> ip 10.24.0.114 255.255.255.240 10.24.0.113
Checking for duplicate address...
PC1 : 10.24.0.114 255.255.255.240 gateway 10.24.0.113

PC4> save
Saving startup configuration to startup.vpc
. done

PC4> show ip

NAME       : PC4[1]
IP/MASK    : 10.24.0.114/28
GATEWAY    : 10.24.0.113
DNS        :
MAC        : 00:50:79:66:68:03
LPORT      : 10034
RHOST:PORT : 127.0.0.1:10035
MTU        : 1500

PC4> █
```

Configuración de interfaces de los routers.

Router 1

```

R1_00004624#
R1_00004624#
R1_00004624#
R1_00004624#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.24.0.1	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	10.24.0.65	YES	manual	up	up
Serial0/1	unassigned	YES	unset	administratively down	down
FastEthernet1/0	10.24.0.129	YES	manual	up	up

```

R1_00004624#

```

Router 2

```

R2_00004624#
R2_00004624#
R2_00004624#
R2_00004624#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.24.0.97	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	10.24.0.130	YES	manual	up	up
Serial0/1	unassigned	YES	unset	administratively down	down
FastEthernet1/0	10.24.0.133	YES	manual	up	up

```

R2_00004624#

```

Router 3

```

R3_00004624#
R3_00004624#
R3_00004624#
R3_00004624#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.24.0.113	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	10.24.0.134	YES	manual	up	up
Serial0/1	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down

```

R3_00004624#

```

Tabla de rutas estáticas configuradas.

Router 1

```

R1_00004624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
C       10.24.0.0/26 is directly connected, FastEthernet0/0
C       10.24.0.64/27 is directly connected, FastEthernet0/1
S       10.24.0.112/28 [1/0] via 10.24.0.130
S       10.24.0.96/28 [1/0] via 10.24.0.130
C       10.24.0.128/30 is directly connected, FastEthernet1/0
S       10.24.0.132/30 [1/0] via 10.24.0.130
R1_00004624#

```

Router 2

```

R2_00004624#
R2_00004624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.24.0.0/26 [1/0] via 10.24.0.129
S       10.24.0.64/27 [1/0] via 10.24.0.129
S       10.24.0.112/28 [1/0] via 10.24.0.134
C       10.24.0.96/28 is directly connected, FastEthernet0/0
C       10.24.0.128/30 is directly connected, FastEthernet0/1
C       10.24.0.132/30 is directly connected, FastEthernet1/0
R2_00004624#

```

Router 3

```

R3_00004624#
R3_00004624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.24.0.0/26 [1/0] via 10.24.0.133
S       10.24.0.64/27 [1/0] via 10.24.0.133
C       10.24.0.112/28 is directly connected, FastEthernet0/0
S       10.24.0.96/28 [1/0] via 10.24.0.133
S       10.24.0.128/30 [1/0] via 10.24.0.133
C       10.24.0.132/30 is directly connected, FastEthernet0/1
R3_00004624#

```

Prueba de trace

```

PC1> ping 10.24.0.114
84 bytes from 10.24.0.114 icmp_seq=1 ttl=61 time=91.428 ms
84 bytes from 10.24.0.114 icmp_seq=2 ttl=61 time=93.531 ms
84 bytes from 10.24.0.114 icmp_seq=3 ttl=61 time=93.499 ms
84 bytes from 10.24.0.114 icmp_seq=4 ttl=61 time=91.906 ms
84 bytes from 10.24.0.114 icmp_seq=5 ttl=61 time=92.959 ms

PC1> trace 10.24.0.114
trace to 10.24.0.114, 8 hops max, press Ctrl+C to stop
 1  10.24.0.1   15.870 ms  15.051 ms  15.131 ms
 2  10.24.0.130 46.218 ms  46.032 ms  46.707 ms
 3  10.24.0.134 76.311 ms  76.521 ms  77.173 ms
 4  *10.24.0.114 90.521 ms (ICMP type:3, code:3, Destination port unreachable)

PC1>

```

DESARROLLO EJERCICIO 2

Carnet: 00004624

A: 00

B: 46

Red base asignada: 192.24.0.0/24

Configuracion Ip de cada PC

```
PC1> ip 192.24.0.2 255.255.255.192 192.24.0.1
Checking for duplicate address...
PC1 : 192.24.0.2 255.255.255.192 gateway 192.24.0.1
```

```
PC1> save
Saving startup configuration to startup.vpc
. done
```

```
PC1> sho ip
```

```
NAME       : PC1[1]
IP/MASK     : 192.24.0.2/26
GATEWAY     : 192.24.0.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 10050
RHOST:PORT  : 127.0.0.1:10051
MTU         : 1500
```

```
PC1> █
```

```
PC2> ip 192.24.0.66 255.255.255.224 192.24.0.65
Checking for duplicate address...
PC1 : 192.24.0.66 255.255.255.224 gateway 192.24.0.65
```

```
PC2> save
Saving startup configuration to startup.vpc
. done
```

```
PC2> show ip
```

```
NAME       : PC2[1]
IP/MASK     : 192.24.0.66/27
GATEWAY     : 192.24.0.65
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 10052
RHOST:PORT  : 127.0.0.1:10053
MTU         : 1500
```

```
PC2> █
```

```
PC3> ip 192.24.0.98 255.255.255.224 192.24.0.97
Checking for duplicate address...
PC1 : 192.24.0.98 255.255.255.224 gateway 192.24.0.97
```

```
PC3> save
Saving startup configuration to startup.vpc
. done
```

```
PC3> show ip
```

```
NAME       : PC3[1]
IP/MASK     : 192.24.0.98/27
GATEWAY     : 192.24.0.97
DNS         :
MAC         : 00:50:79:66:68:02
LPORT      : 10054
RHOST:PORT  : 127.0.0.1:10055
MTU:        : 1500
```

```
PC3> █
```

```
PC4> ip 192.24.0.130 255.255.255.240 192.24.0.129
Checking for duplicate address...
PC1 : 192.24.0.130 255.255.255.240 gateway 192.24.0.129
```

```
PC4> save
Saving startup configuration to startup.vpc
. done
```

```
PC4> show ip
```

```
NAME       : PC4[1]
IP/MASK     : 192.24.0.130/28
GATEWAY     : 192.24.0.129
DNS         :
MAC         : 00:50:79:66:68:03
LPORT      : 10056
RHOST:PORT  : 127.0.0.1:10057
MTU:        : 1500
```

```
PC4> █
```

```

PC5> ip 192.24.0.146 255.255.255.248 192.24.0.145
Checking for duplicate address...
PC1 : 192.24.0.146 255.255.255.248 gateway 192.24.0.145

PC5> save
Saving startup configuration to startup.vpc
. done

PC5> show ip

NAME       : PC5[1]
IP/MASK     : 192.24.0.146/29
GATEWAY     : 192.24.0.145
DNS         :
MAC         : 00:50:79:66:68:04
LPORT      : 10058
RHOST:PORT  : 127.0.0.1:10059
MTU         : 1500

PC5> █

```

Configuración de interfaces de los routers.

Router 1

```

R1_00004624#
R1_00004624#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.0.129   YES manual up          up
Serial0/0       unassigned      YES unset   administratively down down
FastEthernet0/1 192.24.0.153   YES manual up          up
Serial0/1       unassigned      YES unset   administratively down down
FastEthernet1/0 192.24.0.157   YES manual up          up
R1_00004624# █

```

Router 2

```

R2_00004624#
R2_00004624#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.0.65     YES manual up          up
Serial0/0       unassigned      YES unset   administratively down down
FastEthernet0/1 192.24.0.154    YES manual up          up
Serial0/1       unassigned      YES unset   administratively down down
FastEthernet1/0 192.24.0.161    YES manual up          up
R2_00004624# █

```

Router 3

```

R3_00004624#write memory
Building configuration...
[OK]
R3_00004624#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.24.0.97     YES manual up          up
Serial0/0       unassigned      YES unset   administratively down down
FastEthernet0/1 192.24.0.158    YES manual up          up
Serial0/1       unassigned      YES unset   administratively down down
FastEthernet1/0 192.24.0.165    YES manual up          up
R3_00004624# █

```

Router 4

```

R4_00004624#
R4_00004624#
R4_00004624#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          192.24.0.145    YES manual up            up
Serial0/0                 unassigned      YES unset   administratively down down
FastEthernet0/1          192.24.0.166    YES manual up            up
Serial0/1                 unassigned      YES unset   administratively down down
FastEthernet1/0          192.24.0.169    YES manual up            up
R4_00004624#

```

Router 5

```

*Mar 1 00:06:47.591: %SYS-5-CONFIG_1: Configured from console by console
R5_00004624#write memory
Building configuration...
[OK]
R5_00004624#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          192.24.0.1      YES manual up            up
Serial0/0                 unassigned      YES unset   administratively down down
FastEthernet0/1          192.24.0.162    YES manual up            up
Serial0/1                 unassigned      YES unset   administratively down down
FastEthernet1/0          192.24.0.170    YES manual up            up
R5_00004624#

```

Configuración OSPF. Captura de la tabla de rutas.

Router 1

```

R1_00004624#
R1_00004624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 5 masks
C       192.24.0.152/30 is directly connected, FastEthernet0/1
C       192.24.0.156/30 is directly connected, FastEthernet1/0
O       192.24.0.144/29 [110/12] via 192.24.0.158, 00:00:34, FastEthernet1/0
C       192.24.0.128/28 is directly connected, FastEthernet0/0
O       192.24.0.168/30 [110/3] via 192.24.0.158, 00:00:34, FastEthernet1/0
O       192.24.0.160/30 [110/11] via 192.24.0.154, 00:05:45, FastEthernet0/1
O       192.24.0.164/30 [110/2] via 192.24.0.158, 00:04:00, FastEthernet1/0
O       192.24.0.64/27 [110/20] via 192.24.0.154, 00:05:57, FastEthernet0/1
O       192.24.0.96/27 [110/11] via 192.24.0.158, 00:04:10, FastEthernet1/0
O       192.24.0.0/26 [110/13] via 192.24.0.158, 00:00:36, FastEthernet1/0
R1_00004624#

```

Router 2


```

R2_00004624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 5 masks
C       192.24.0.152/30 is directly connected, FastEthernet0/1
O       192.24.0.156/30 [110/11] via 192.24.0.153, 00:05:47, FastEthernet0/1
O       192.24.0.144/29 [110/12] via 192.24.0.162, 00:00:28, FastEthernet1/0
O       192.24.0.128/28 [110/20] via 192.24.0.153, 00:05:47, FastEthernet0/1
O       192.24.0.168/30 [110/2] via 192.24.0.162, 00:02:10, FastEthernet1/0
C       192.24.0.160/30 is directly connected, FastEthernet1/0
O       192.24.0.164/30 [110/12] via 192.24.0.162, 00:00:30, FastEthernet1/0
        [110/12] via 192.24.0.153, 00:03:55, FastEthernet0/1
C       192.24.0.64/27 is directly connected, FastEthernet0/0
O       192.24.0.96/27 [110/21] via 192.24.0.153, 00:03:55, FastEthernet0/1
O       192.24.0.0/26 [110/11] via 192.24.0.162, 00:02:22, FastEthernet1/0
R2_00004624#

```

Router 3

```

R3_00004624#
R3_00004624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 5 masks
O       192.24.0.152/30 [110/20] via 192.24.0.157, 00:03:44, FastEthernet0/1
C       192.24.0.156/30 is directly connected, FastEthernet0/1
O       192.24.0.144/29 [110/11] via 192.24.0.166, 00:00:24, FastEthernet1/0
O       192.24.0.128/28 [110/20] via 192.24.0.157, 00:03:44, FastEthernet0/1
O       192.24.0.168/30 [110/2] via 192.24.0.166, 00:00:14, FastEthernet1/0
O       192.24.0.160/30 [110/12] via 192.24.0.166, 00:00:14, FastEthernet1/0
C       192.24.0.164/30 is directly connected, FastEthernet1/0
O       192.24.0.64/27 [110/22] via 192.24.0.166, 00:00:16, FastEthernet1/0
C       192.24.0.96/27 is directly connected, FastEthernet0/0
O       192.24.0.0/26 [110/12] via 192.24.0.166, 00:00:16, FastEthernet1/0
R3_00004624#

```

Router 4

```

R4_00004624#
R4_00004624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 5 masks
O       192.24.0.152/30 [110/21] via 192.24.0.170, 00:00:07, FastEthernet1/0
O       192.24.0.156/30 [110/20] via 192.24.0.165, 00:00:07, FastEthernet0/1
C       192.24.0.144/29 is directly connected, FastEthernet0/0
O       192.24.0.128/28 [110/30] via 192.24.0.165, 00:00:07, FastEthernet0/1
C       192.24.0.168/30 is directly connected, FastEthernet1/0
O       192.24.0.160/30 [110/11] via 192.24.0.170, 00:00:07, FastEthernet1/0
C       192.24.0.164/30 is directly connected, FastEthernet0/1
O       192.24.0.64/27 [110/21] via 192.24.0.170, 00:00:09, FastEthernet1/0
O       192.24.0.96/27 [110/20] via 192.24.0.165, 00:00:09, FastEthernet0/1
O       192.24.0.0/26 [110/11] via 192.24.0.170, 00:00:09, FastEthernet1/0
R4_00004624#

```

Router 5

```

R5_00004624#
R5_00004624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 5 masks
O       192.24.0.152/30 [110/20] via 192.24.0.161, 00:02:04, FastEthernet0/1
O       192.24.0.156/30 [110/21] via 192.24.0.169, 00:00:20, FastEthernet1/0
       [110/21] via 192.24.0.161, 00:02:04, FastEthernet0/1
O       192.24.0.144/29 [110/11] via 192.24.0.169, 00:00:20, FastEthernet1/0
O       192.24.0.128/28 [110/30] via 192.24.0.161, 00:02:04, FastEthernet0/1
C       192.24.0.168/30 is directly connected, FastEthernet1/0
C       192.24.0.160/30 is directly connected, FastEthernet0/1
O       192.24.0.164/30 [110/11] via 192.24.0.169, 00:00:22, FastEthernet1/0
O       192.24.0.64/27 [110/20] via 192.24.0.161, 00:02:06, FastEthernet0/1
O       192.24.0.96/27 [110/21] via 192.24.0.169, 00:00:22, FastEthernet1/0
C       192.24.0.0/26 is directly connected, FastEthernet0/0
R5_00004624#

```

Prueba de trace

```
PC5> ping 192.24.0.130
84 bytes from 192.24.0.130 icmp_seq=1 ttl=61 time=77.745 ms
84 bytes from 192.24.0.130 icmp_seq=2 ttl=61 time=93.607 ms
84 bytes from 192.24.0.130 icmp_seq=3 ttl=61 time=93.073 ms
84 bytes from 192.24.0.130 icmp_seq=4 ttl=61 time=78.423 ms
84 bytes from 192.24.0.130 icmp_seq=5 ttl=61 time=91.113 ms

PC5> trace 192.24.0.130
trace to 192.24.0.130, 8 hops max, press Ctrl+C to stop
 1  192.24.0.145    15.086 ms  12.815 ms  14.593 ms
 2  192.24.0.165    46.039 ms  46.138 ms  47.064 ms
 3  192.24.0.157    78.091 ms  77.265 ms  62.184 ms
 4  *192.24.0.130   75.990 ms (ICMP type:3, code:3, Destination port unreachable)

PC5> █
```